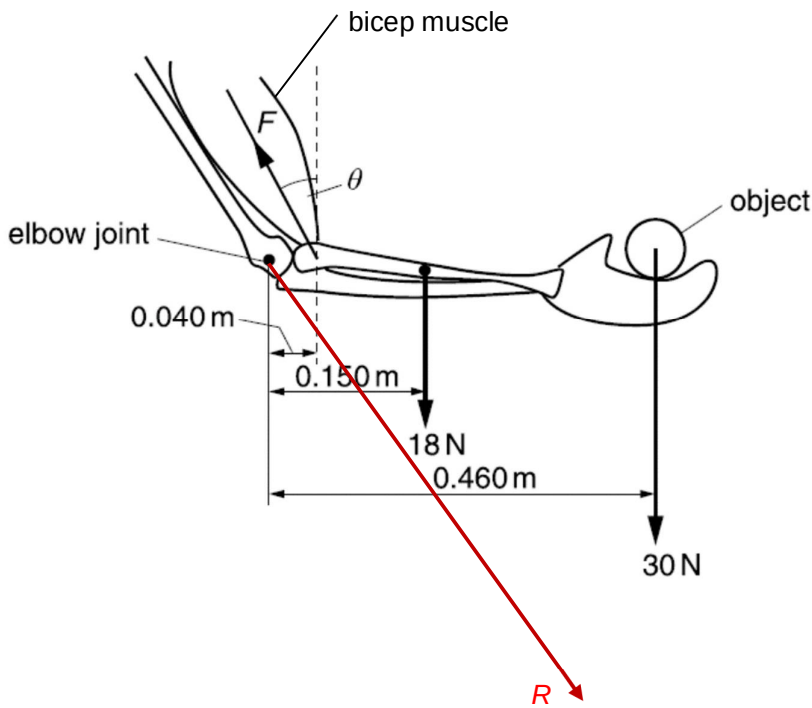
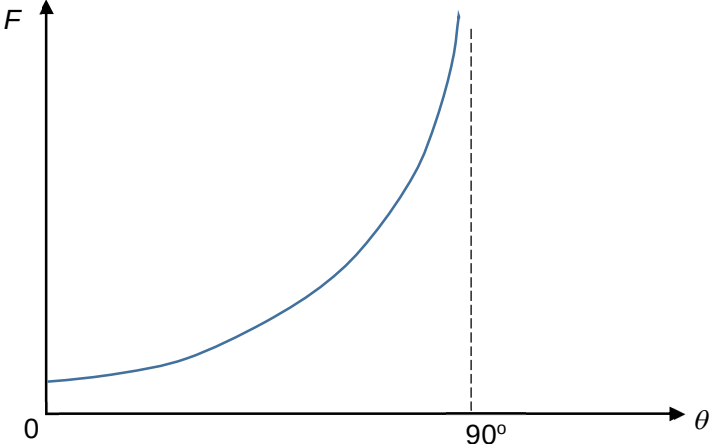


Qn 1	Answer	Marks
(a)	It is the <u>point where the weight appears to act</u> . Do not accept: "The point where gravity acts" or "point where mass acts/is concentrated"	B1
(b)	Taking moments about the elbow joint, $(F \cos 15.0^\circ)(0.040) = (18)(0.150) + (30)(0.460)$ $\Rightarrow F = 427 \text{ N}$	M1 A1
(c)(i)	 Correct direction of R.	A1
(c)(ii)	'The rightward horizontal component of R is to balance the leftward horizontal component of F. <u>Taking moments about the point the muscle is attached to the bone</u>, the vertical component of R needs to act downwards to provide an anticlockwise moment to counter the clockwise moment provided by the 18 N and 30 N forces.' OR 'The rightward horizontal component of R is to balance the leftward horizontal component of F. The net vertical force due to the weights and tension is upwards, hence the vertical component of R must act downwards so that there is no net vertical force.' OR 'A vector diagram is draw in scale. Using drawn vector diagram in scale, we can deduce the direction of the force R should point rightward and downward.' OR 'The forces of 18 N and 30 N can be combined to form one downwards force which should be drawn between the two mentioned forces. Since the forces of R, F and combined force due to 18 N and 30 N should intersect at a point, we can deduce the direction of the force R should point leftward and downward.'	B1 B1 B1 B1 B1 B1 B1 B1

(d)(i)	 Curve is sloping upwards and does not start from origin, where its asymptote is at 90°	B1 B1
(d)(ii)	Consider moments about the elbow joint, The perpendicular distance between the elbow and the line of action of F decreases or the clockwise moment due to the 18 N and 30 N forces remain the same and $F \cos \theta$ (or vertical component of F) hence remains the constant. Hence, as θ increase, ($\cos \theta$ reduces), F increases. OR F is inversely proportional to $\cos \theta$. When θ approaches 90°, F is infinitely large	A1 A1
	Max Marks	10

Question	Answer	Marks
2(a)	The internal energy is the summation of microscopic kinetic energy due to random motion of the	A2